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Post-Flood Sediment Deposition Analysis Using SAR Interferometry: A Case Study of the 2024 Flood in Feni district, Bangladesh

Abdullah Al Jahib, Md Asadullahil Galib Fardin, and Sameer Mahmud Khan
Institute of Water and Flood Management (IWFM), Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh (jahib114@gmail.com)

Being a country with heavy monsoon rains and seasonal flooding, every year Bangladesh faces severe damage, some of which are irreparable. The flood of 2024 which occurred in August, severely affected the north-eastern and south-eastern part of the country. Feni district was severely affected because of heavy water flow from the Dumboor dam in Tripura state. Subsequently, bringing massive amount of sediment load through the hilly area. These flooding events cause significant damage to human life in the region, the flood-associated sediment is also crucial for the region's agriculture. Therefore, the purpose of this study was to find the amount of sediment deposited due to the flood in the Feni floodplain. To achieve this objective, interferometry methods were used to create the Digital Elevation Models (DEMs) of the floodplains before and after the flood. To analyze the flood-plain state before the flood, two images were selected from 29th May to 10th June and two images were selected after the flood between 8th to 20th October. Single Look Complex (SLC) products were chosen, which consist of focused SAR data, geo-referenced using orbit , attitude data from the satellite and provided in slant-range geometry. The Sentinel-1 images were pre-processed . Than phase was unwrapped, and converted to DEM. The software SNAP, which is a common architecture of all Sentinel toolboxes, was used for DEM generation. After the DEM generation, 8 profiles were drawn to observe the change in topography due to sedimentation by HEC-RAS software. It was found that there was an average of 2.7 cm sediment deposition throughout the region and a 7.5 cm maximum deposition.